

Nour Belgacem

Electrical Engineer

C / C++ / Qt / Embedded Systems

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Nour Belgacem
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PROFESSIONAL EXPERIENCE

- **Development of Multi-Protocol Interfaces for a Test Bench in Qt/C++** *February 2026 – July 2026*

  Final-Year Engineering Internship SERMA International, Tunisia

  - Implemented the CAN, CAN FD and RS-232/UART protocols for memory data acquisition (RAM/FRAM) and real-time communication with embedded equipment.
  - Developed Qt/C++ graphical interfaces achieving sub-millisecond timing accuracy in data exchange over these protocols.
  - Eliminated a latency issue in the CAN interface by designing an IPC-based architecture, with dedicated transmission and reception worker processes separated from the GUI process.
  - Validated the CAN interface on a hardware setup built around an STM32 board and a CAN module.
  - Built a qualification interface for aeronautical equipment, integrating signal acquisition through an NI USB-6001 board and automatic test report generation.
  - Engineered a driver for a programmable power supply (DPX400) from its programming manual, to set and read the measured voltage and current values.
  - Verified the interface tests against aeronautical standards and protocols (DO-160).

  Key skills: CAN/CAN FD, RS-232/UART, Qt/C++, IPC, Multiprocess architecture, STM32, Driver development, NI USB-6001, DO-160, Embedded systems, Aeronautics, Scrumban, Jira.

PROJECTS

- **Industrial Supervision Dashboard (SupervisionDashboard)** *2026*

  Personal Project Qt 6 / QML / C++17

  - Developed a C++17 and Qt 6/QML desktop application to monitor simulated electrical parameters in real time, including voltage, current, power, and temperature.
  - Implemented MQTT communication between a Python-based data simulator running on Linux and the Qt application to publish and receive simulated sensor measurements.
  - Integrated real-time charts, a temperature gauge, warning/critical alarms, and light/dark themes, ensuring clear separation between the C++ backend and QML interface.
  - Integrated SQLite history with CSV export and persistent settings. Managed the project with Git/GitHub and CMake.
  - Wrote unit tests with Qt Test to validate the software logic and ensure software quality assurance, and verified the interface behavior through the automatically generated pass/fail results.

  Key skills: C++17, Qt 6, QML, MQTT, Python, Linux, Qt Graphs, Qt Test, Software Quality Assurance, CMake, Git/GitHub, SQLite, Real-Time Monitoring.

- **Autonomous Fire Fighting Robot with RTOS** *September 2025 – January 2026*

  Academic Project ENIT, Tunisia

  - Developed embedded software in C to detect fire and control an automated water-pump extinguishing system using an STM32 microcontroller.
  - Implemented FreeRTOS task scheduling and priority management to coordinate sensor monitoring, robot movement, and fire-extinguishing operations.
  - Designed a Qt/C++ graphical user interface (GUI) for serial communication, enabling user interaction and robot monitoring.

  Key skills: Embedded C, STM32F407, STM32CubeIDE, FreeRTOS,RTOS,Task Scheduling,UART/Serial.

EDUCATION

- **National Engineering School of Tunis (ENIT)** *September 2023 – June 2026*

  Engineering Degree in Electrical Engineering – *Advanced Electronic Systems (SEA)* Tunis, Tunisia

CERTIFICATIONS

- **C++ Programming:** Saylor Academy *(Certificate ID: 6074865565NB)*
- **Prompt Engineering and Programming with OpenAI:** Columbia + *(Certificate ID: 197332618)*

TECHNICAL SKILLS

- **Programming languages:** C, C++17,Python, QML
- **Software and embedded development:** Qt Creator, Qt Framework, STM32CubeIDE, VS Code, Embadded Linux

LANGUAGES

French: B2 (DELF) **English:** B2 (TOEIC: 785)